GATE ECE 2027 -- TARGET-BASED DAILY QUEST CHECKLIST

THE "RULE OF 3" DAILY QUEST: No timestamps. Work whenever you want. Check 3 boxes -> Day is WON.

[ ] 1. Theory Input: 1 Topic Module | [ ] 2. Active Output: 15-20 PYQs (4-min timer) | [ ] 3. Retention: 1-Page Note + Error Log

STAGE 1: THE UNSHAKABLE FOUNDATION (WEEKS 1-4) — TARGET: 28 MARKS LOCKED

[ ] D1	Math: Lin Alg	Matrices, Determinant shortcuts, Row Echelon Rank	15 PYQs	1-Page Note
[ ] D2	Math: Lin Alg	Systems Ax=b consistency (Rank(A) vs Rank[A|b])	15 PYQs	Error Log
[ ] D3	Math: Lin Alg	Eigenvalues & Eigenvectors (Sum=Tr, Prod=Det)	20 PYQs	Error Log
[ ] D4	Math: Lin Alg	Cayley-Hamilton Theorem & Quadratic Forms	20 PYQs	1-Page Note
[ ] D5	Math: Lin Alg	Vector Spaces, Basis, Rank-Nullity Theorem + GA Verbal	20 PYQs	Error Log
[ ] D6	**TEST DAY**	**Subject Test 1: Linear Algebra + GA Verbal (30M)**	**CBT Test**	**5-Step Analysis**
[ ] D7	**AUDIT & REST**	**Full-Week Blank-Sheet Memory Dump + Red Star Rework**	**--**	**Mandatory Rest**
[ ] D8	Math: Calculus	Limits (L'Hopital, 1^inf), Continuity, Mean Value Thms	15 PYQs	1-Page Note
[ ] D9	Math: Calculus	Maxima & Minima (rt-s^2 test), Taylor Series	15 PYQs	Error Log
[ ] D10	Math: Calculus	Definite Integrals, Leibniz differentiation rule	15 PYQs	Error Log
[ ] D11	Math: Vector Calc	Gradient, Directional Derivative, Divergence, Curl	20 PYQs	1-Page Note
[ ] D12	Math: Vector Calc	Vector Theorems (Gauss, Green, Stokes) + GA Quant 1	20 PYQs	Error Log
[ ] D13	**TEST DAY**	**Subject Test 2: Calculus + GA Quant (30M)**	**CBT Test**	**5-Step Analysis**
[ ] D14	**AUDIT & REST**	**Calculus Formula Memory Dump + Error Log Audit**	**--**	**Mandatory Rest**
[ ] D15	Math: ODEs	1st-Order ODEs (Separable, Exact, Integrating factor)	15 PYQs	1-Page Note
[ ] D16	Math: ODEs	Higher-Order Linear ODEs (CF + PI, Cauchy-Euler)	20 PYQs	Error Log
[ ] D17	Math: Probability	Conditional Probability & Bayes' Theorem	20 PYQs	Error Log
[ ] D18	Math: Probability	Random Variables: PDF, CDF, Uniform & Exponential	20 PYQs	1-Page Note
[ ] D19	Math: Probability	Normal Distribution, Binomial, Poisson + GA P&C	20 PYQs	Error Log
[ ] D20	**TEST DAY**	**Subject Test 3: Diff Eq & Probability (30M)**	**CBT Test**	**5-Step Analysis**
[ ] D21	**AUDIT & REST**	**Probability Axioms Recall + Error Log Audit**	**--**	**Mandatory Rest**
[ ] D22	Math: Complex	Cauchy-Riemann (C-R) Equations, Analytic functions	15 PYQs	1-Page Note
[ ] D23	Math: Complex	Complex Integrals & Cauchy Integral Formula	15 PYQs	Error Log
[ ] D24	Math: Complex	Singularities, Laurent Series & Cauchy Residue Theorem	20 PYQs	1-Page Note
[ ] D25	Math: PDE / GA	PDE Separation of Variables + GA Spatial/DI Puzzles	20 PYQs	Error Log
[ ] D26	Math: Master Rev	Math Formula Master Binder Speed-Run (All 16 Pages)	25 PYQs	Formula Consolidation
[ ] D27	**STAGE 1 TEST**	**STAGE 1 SIMULATION TEST (Full Math + GA = 28M CBT)**	**28M Test**	**Stage 1 Audit**
[ ] D28	**STAGE 1 LOCK**	**Stage 1 Victory Audit (Target: >= 24/28 Marks Locked)**	**--**	**STAGE 1 COMPLETE**

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STAGE 2A: CIRCUITS & DETERMINISTIC LOGIC (WEEKS 5-7)				+14 MARKS (44M CUMULATIVE)
[ ] D29	Networks: Laws	KVL/KCL, Mesh/Nodal with Dependent Sources	15 PYQs	1-Page Note
[ ] D30	Networks: Theorems	Thevenin & Norton with Dependent Sources	20 PYQs	Error Log
[ ] D31	Networks: Theorems	Maximum Power Transfer (Z_L = Z_th*) & Superposition	15 PYQs	Error Log
[ ] D32	Digital: K-Maps	Boolean Minimization & K-Maps with Don't-Cares	20 PYQs	1-Page Note
[ ] D33	Digital: Comb.	Multiplexers (MUX logic), Decoders, Encoders, Adders	20 PYQs	Error Log
[ ] D34	**TEST DAY**	**Subject Test 4: Networks 1 + Digital 1 (30M)**	**CBT Test**	**5-Step Analysis**
[ ] D35	**AUDIT & REST**	**Circuit Laws Blank Sheet Recall + Error Review**	**--**	**Mandatory Rest**
[ ] D36	Networks: Transients	1st-Order RL/RC Transients: v(t)=v(inf)+[v(0+)-v(inf)]e^(-t/tau)	20 PYQs	1-Page Note
[ ] D37	Networks: Transients	2nd-Order RLC Transients & AC Phasor Analysis	15 PYQs	Error Log
[ ] D38	Networks: Resonance	Series & Parallel Resonance (w0=1/sqrt(LC), Q, BW)	20 PYQs	1-Page Note
[ ] D39	Digital: Flip-Flops	Flip-Flops (D, T, JK), Characteristic & Excitation Tables	20 PYQs	Error Log
[ ] D40	Digital: Counters	Synchronous & Ripple Modulo-N Counters (fmax=1/Ntpd)	20 PYQs	1-Page Note
[ ] D41	**TEST DAY**	**Subject Test 5: Transients + Counters (30M)**	**CBT Test**	**5-Step Analysis**
[ ] D42	**AUDIT & REST**	**Transients & Counters Memory Dump + Error Log**	**--**	**Mandatory Rest**
[ ] D43	Networks: 2-Port	Two-Port Parameters (Z, Y, h, ABCD) & Conversions	15 PYQs	1-Page Note
[ ] D44	Networks: 2-Port	Two-Port Interconnections, Reciprocity & Symmetry	15 PYQs	Error Log
[ ] D45	Digital: FSM	Finite State Machines: Moore vs Mealy State Tables	20 PYQs	1-Page Note
[ ] D46	Digital: Timing	Digital Timing Closure: Setup (t_setup) & Hold (t_hold) Slack	20 PYQs	Error Log
[ ] D47	Digital: CMOS	CMOS Inverter Logic, Dynamic Power (CL*Vdd^2*f), Fan-out	20 PYQs	1-Page Note
[ ] D48	**STAGE 2A TEST**	**SECTIONAL TEST: FULL NETWORKS + FULL DIGITAL (50M)**	**50M Test**	**Stage 2A Audit**
[ ] D49	**STAGE 2A LOCK**	**Stage 2A Victory Audit (Cumulative: 44 Marks Locked)**	**--**	**STAGE 2A COMPLETE**

STAGE 2B: TRANSFORMS & SOLID-STATE PHYSICS (WEEKS 8-11)				+13 MARKS (56M CUMULATIVE)
[ ] D50	Signals: CT Basics	Signal Operations, Energy vs Power, Singularities	15 PYQs	1-Page Note
[ ] D51	Signals: LTI	LTI Systems, BIBO Stability (int |h(t)|dt < inf)	15 PYQs	Error Log
[ ] D52	Signals: LTI	Convolution Integral (Graphical flip-and-slide)	20 PYQs	1-Page Note
[ ] D53	EDC: Physics	Semiconductor Bands, Carrier Density (np=ni^2), Fermi	15 PYQs	1-Page Note
[ ] D54	EDC: Physics	Carrier Transport: Drift, Diffusion, Einstein Rel (D/mu=Vt)	20 PYQs	Error Log
[ ] D55	**TEST DAY**	**Subject Test 6: CT Signals + Semiconductor Physics (30M)**	**CBT Test**	**5-Step Analysis**
[ ] D56	**AUDIT & REST**	**Signals & EDC Memory Dump + Error Log Audit**	**--**	**Mandatory Rest**

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[ ] D57	Signals: Fourier	Continuous-Time Fourier Series (CTFS) & Parseval Power	15 PYQs	1-Page Note
[ ] D58	Signals: Fourier	Continuous-Time Fourier Transform (CTFT) Standard Pairs	15 PYQs	Error Log
[ ] D59	Signals: Fourier	CTFT Properties: Duality, Modulation, Parseval Energy	20 PYQs	1-Page Note
[ ] D60	EDC: P-N Diode	P-N Junction: Built-in Potential (Vbi), Depletion Width (W)	20 PYQs	1-Page Note
[ ] D61	EDC: P-N Diode	Diode Dynamics: Cj proportional to VR^(-1/2), Cd, Zener	20 PYQs	Error Log
[ ] D62	**TEST DAY**	**Subject Test 7: Fourier Transforms + Diodes (30M)**	**CBT Test**	**5-Step Analysis**
[ ] D63	**AUDIT & REST**	**Fourier & Diode Memory Dump + Error Review**	**--**	**Mandatory Rest**
[ ] D64	Signals: Laplace	Bilateral Laplace Transform & ROC Properties	15 PYQs	1-Page Note
[ ] D65	Signals: Laplace	Inverse Laplace, Transfer Function Stability, Initial/Final	15 PYQs	Error Log
[ ] D66	Signals: Sampling	Sampling Theorem: Nyquist Rate (2fm), Aliasing, Sinc filter	20 PYQs	1-Page Note
[ ] D67	EDC: MOS Cap	MOS Capacitor: Band Bending, Inversion, C-V Curves	15 PYQs	1-Page Note
[ ] D68	EDC: MOSFET	MOSFET Physics: Triode vs Saturation ID, Vth, Body Effect	25 PYQs	1-Page Note
[ ] D69	**TEST DAY**	**Subject Test 8: Laplace + MOSFET Physics (30M)**	**CBT Test**	**5-Step Analysis**
[ ] D70	**AUDIT & REST**	**Laplace & MOSFET Recall + Error Review**	**--**	**Mandatory Rest**
[ ] D71	Signals: DTFT	DTFT Properties & Discrete-Time LTI Frequency Response	15 PYQs	1-Page Note
[ ] D72	Signals: Z-Trans	Z-Transform ROC Properties & Stability (|z|=1 inclusion)	20 PYQs	1-Page Note
[ ] D73	Signals: Z-Trans	Inverse Z-Transform via Partial Fractions, Diff Equations	20 PYQs	Error Log
[ ] D74	EDC: BJT Physics	BJT Physics: Active/Sat/Cutoff, Early Effect (VA)	20 PYQs	1-Page Note
[ ] D75	EDC: BJT Model	BJT Small-Signal pi-Model (gm, rpi, ro, fT)	20 PYQs	1-Page Note
[ ] D76	**STAGE 2B TEST**	**SECTIONAL TEST: FULL SIGNALS + FULL EDC (60M CBT)**	**60M Test**	**Stage 2B Audit**
[ ] D77	**STAGE 2B LOCK**	**Stage 2B Victory Audit (Cumulative: 56 Marks Locked)**	**--**	**STAGE 2B COMPLETE**

STAGE 3: AMPLIFIERS, CONTROL & SYNTHESIS (WEEKS 12-15)				**+12 MARKS (68M CUMULATIVE)**
[ ] D78	Analog: Op-Amps	Ideal Op-Amp Model, Virtual Ground, Inverting/Non-Inv	15 PYQs	1-Page Note
[ ] D79	Analog: Op-Amps	Linear Op-Amps: Summing, Difference, Integrator	20 PYQs	Error Log
[ ] D80	Analog: Op-Amps	Non-Linear Op-Amps: Comparators, Schmitt Trigger	20 PYQs	1-Page Note
[ ] D81	Control: Basics	Transfer Functions, Block Diagrams & Mason's Gain Formula	20 PYQs	1-Page Note
[ ] D82	Control: Basics	Standard Test Signals, Poles/Zeros on s-plane	15 PYQs	Error Log
[ ] D83	**TEST DAY**	**Subject Test 9: Op-Amps + Mason's Gain (30M)**	**CBT Test**	**5-Step Analysis**
[ ] D84	**AUDIT & REST**	**Op-Amps & Control Basics Recall + Error Review**	**--**	**Mandatory Rest**

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[ ] D85	Analog: BJT Amps	BJT Biasing, CE Amplifier Small-Signal Gain & Impedances	15 PYQs	1-Page Note
[ ] D86	Analog: BJT Amps	BJT CC (Emitter Follower) & CB Amplifiers	15 PYQs	Error Log
[ ] D87	Analog: MOS Amps	MOSFET CS (Common Source) & CD (Source Follower) Amps	20 PYQs	1-Page Note
[ ] D88	Control: Time Resp	2nd-Order Systems: Characteristic Equation, zeta, wn	20 PYQs	1-Page Note
[ ] D89	Control: Time Resp	Transient Specs (Mp, tp, ts) & Steady-State Error (Kp, Kv, Ka)	20 PYQs	Error Log
[ ] D90	**TEST DAY**	**Subject Test 10: Amplifiers + 2nd Order Control (30M)**	**CBT Test**	**5-Step Analysis**
[ ] D91	**AUDIT & REST**	**Amplifiers & Time Response Recall + Error Review**	**--**	**Mandatory Rest**
[ ] D92	Analog: Feedback	4 Feedback Topologies (Identification & Gain/Impedance rules)	20 PYQs	1-Page Note
[ ] D93	Analog: Diodes	Diode Circuits: Clippers, Clampers, Peak-to-Peak Conservation	20 PYQs	1-Page Note
[ ] D94	Control: Routh	Routh-Hurwitz Stability Criterion (Special Cases & Range of K)	20 PYQs	1-Page Note
[ ] D95	Control: Root Locus	Root Locus Construction Rules (Asymptotes, Centroid, Breakaway)	20 PYQs	1-Page Note
[ ] D96	Control: Root Locus	Root Locus Angle of Departure/Arrival, Stability via RL	15 PYQs	Error Log
[ ] D97	**TEST DAY**	**Subject Test 11: Feedback + Root Locus (30M)**	**CBT Test**	**5-Step Analysis**
[ ] D98	**AUDIT & REST**	**Feedback & Root Locus Memory Dump + Error Review**	**--**	**Mandatory Rest**
[ ] D99	Analog: Oscillators	Barkhausen Criterion, RC Phase Shift & Wien Bridge	15 PYQs	1-Page Note
[ ] D100	Analog: 555 Timer	555 Timer IC (Astable & Monostable Multivibrator formulas)	15 PYQs	Error Log
[ ] D101	Control: Bode Plot	Bode Plots: GM, PM, Asymptotic Transfer Function Extraction	25 PYQs	1-Page Note
[ ] D102	Control: Nyquist	Nyquist Stability Criterion (N = P - Z Enclosure Rule)	20 PYQs	1-Page Note
[ ] D103	Control: State Space	State Space: State Transition Matrix (e^At), Kalman Tests	20 PYQs	1-Page Note
[ ] D104	**STAGE 3 TEST**	**SECTIONAL TEST: FULL ANALOG + FULL CONTROL (60M CBT)**	**60M Test**	**Stage 3 Audit**
[ ] D105	**STAGE 3 LOCK**	**Stage 3 Victory Audit (Cumulative: 68 Marks Locked)**	**--**	**STAGE 3 COMPLETE**
STAGE 4: ADVANCED FRONTIER & FINISH-LINE (WEEKS 16-18)				**+15 MARKS (~72-74M NET)**
[ ] D106	Comms: Random Proc	Wide-Sense Stationary (WSS), Autocorrelation (R_XX)	15 PYQs	1-Page Note
[ ] D107	Comms: Noise	Power Spectral Density (S_XX) & Noise through LTI Filters	20 PYQs	1-Page Note
[ ] D108	Comms: Noise Fig	Noise Figure (F) & Friis Cascaded Noise Formula	15 PYQs	Error Log
[ ] D109	EMTL: T-Lines	Transmission Lines (70% of EMTL): Z0, Gamma, VSWR	20 PYQs	1-Page Note
[ ] D110	EMTL: T-Lines	Quarter-Wave Transformer (Zin = Z0^2/ZL) & Matching	20 PYQs	Error Log
[ ] D111	**TEST DAY**	**Subject Test 12: Random Processes + T-Lines (30M)**	**CBT Test**	**5-Step Analysis**
[ ] D112	**AUDIT & REST**	**Noise & T-Lines Memory Dump + Error Review**	**--**	**Mandatory Rest**
[ ] D113	Comms: Analog Mod	AM Modulation (DSB-FC power Pt, DSB-SC, SSB Hilbert)	20 PYQs	1-Page Note
[ ] D114	Comms: Angle Mod	FM Angle Modulation & Carson's Bandwidth Rule (2(Df+fm))	20 PYQs	1-Page Note
[ ] D115	Comms: PCM	Pulse Code Modulation: Quantization Noise (Delta^2/12), SNRq	20 PYQs	1-Page Note
[ ] D116	EMTL: Waves	Maxwell's Equations & Uniform Plane Waves in Media	15 PYQs	1-Page Note
[ ] D117	EMTL: Waves	Skin Depth (delta), Poynting Vector (P_avg), Wave Polarization	20 PYQs	Error Log
[ ] D118	**TEST DAY**	**Subject Test 13: Analog Comms + Plane Waves (30M)**	**CBT Test**	**5-Step Analysis**
[ ] D119	**AUDIT & REST**	**AM/FM & Plane Waves Recall + Error Review**	**--**	**Mandatory Rest**
[ ] D120	Comms: Digital Mod	BPSK, QPSK, 16-QAM Constellations & Bit Error Rate (BER)	25 PYQs	1-Page Note
[ ] D121	Comms: Detection	Matched Filter Impulse Response (h(t)=s*(T-t)) & ML Detection	20 PYQs	1-Page Note
[ ] D122	Comms: Info Theory	Information Theory: Entropy (H), Shannon Channel Capacity	20 PYQs	1-Page Note
[ ] D123	EMTL: Waveguides	Rectangular Waveguides: Dominant Mode (TE10), fc, lambda_g	20 PYQs	1-Page Note
[ ] D124	EMTL: Antennas	Antennas: Directivity (D), Gain, Friis Transmission Formula	15 PYQs	Error Log
[ ] D125	**STAGE 4 TEST**	**SECTIONAL TEST: FULL COMMS + FULL EMTL (60M CBT)**	**60M Test**	**Stage 4 Audit**
[ ] D126	**STAGE 4 LOCK**	**GRAND DECEMBER SYLLABUS COMPLETION AUDIT**	**--**	**100% SYLLABUS DONE**